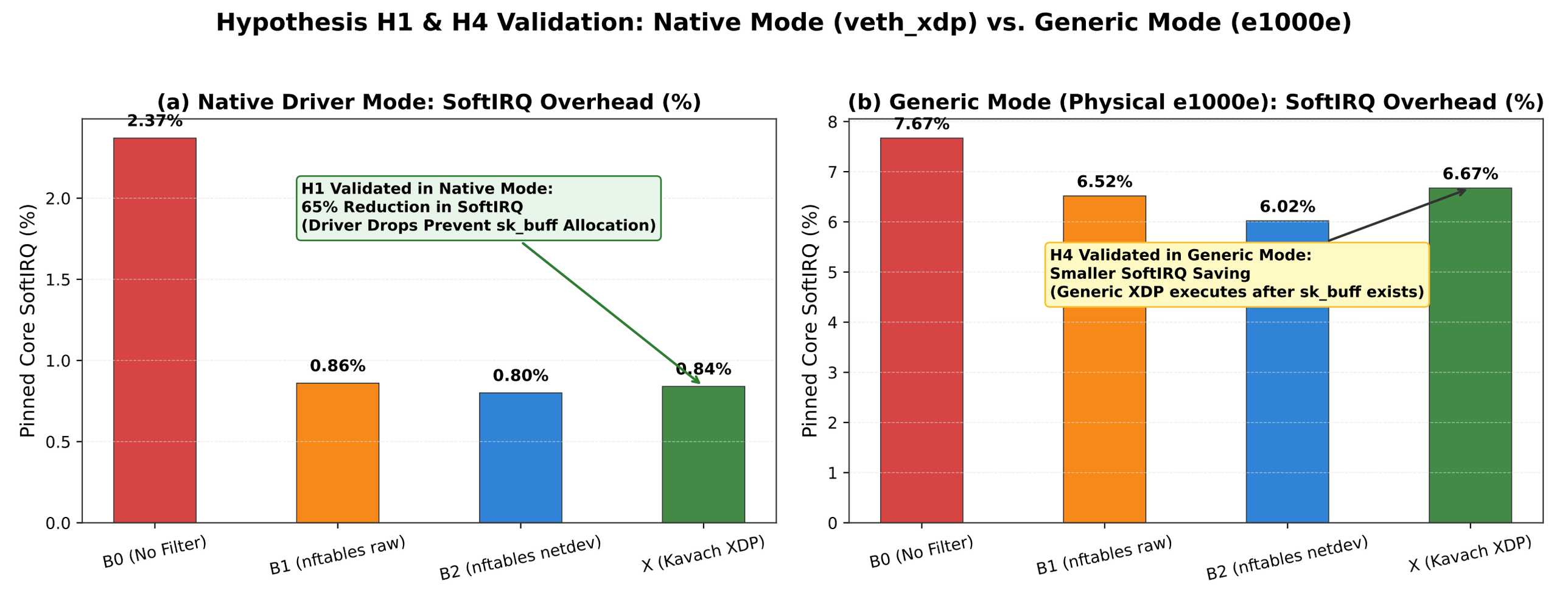
Hypothesis H1 & H4 Validation: Native Mode (veth_xdp) vs. Generic Mode (e1000e)
(a) Native Driver Mode: SoftIRQ Overhead (%)
Pinned Core SoftIRQ (%)
2.37%
0.86%
0.80%
0.84%
H1 Validated in Native Mode:
65% Reduction in SoftIRQ
(Driver Drops Prevent sk_buff Allocation)
B0 (No Filter)
B1 (nftables raw)
B2 (nftables netdev)
X (Kavach XDP)
(b) Generic Mode (Physical e1000e): SoftIRQ Overhead (%)
Pinned Core SoftIRQ (%)
7.67%
6.52%
6.02%
6.67%
H4 Validated in Generic Mode:
Smaller SoftIRQ Saving
(Generic XDP executes after sk_buff exists)
B0 (No Filter)
B1 (nftables raw)
B2 (nftables netdev)
X (Kavach XDP)